Compton scattering 4

From the first section we have

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(s-m^2)^2} \left(-8su + 24m^2 s + 8m^2 u + 8m^4\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(s-m^2)(u-m^2)} \left(8m^2 s + 8m^2 u + 16m^4\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left(-8su + 8m^2 s + 24m^2 u + 8m^4\right)$$

We also have by computer algebra

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(s-m^2)^2} \left(32(p_1 \cdot p_2)(p_1 \cdot p_4) + 32m^2(p_1 \cdot p_2) + 32m^4\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(s-m^2)(u-m^2)} \left(16m^2(p_1 \cdot p_2) - 16m^2(p_1 \cdot p_4) + 32m^4\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left(32(p_1 \cdot p_2)(p_1 \cdot p_4) - 32m^2(p_1 \cdot p_4) + 32m^4\right)$$

Eigenmath script